

NMTC SYLLABUS

SYLLABUS:

Recommended topics for NMTC exam

PRIMARY:

(Gauss Contest - V and VI Standards)

ARITHMETIC: Fractions, percentages, profit and loss, tests of divisibility, LCM, HCF, Ratio and proportion, Calendar

MENSURATION: Triangles, Quadrilaterals, Circles.

ALGEBRA: Algebra as literal Arithmetic, Addition, subtraction, multiplication and division of simple Arithmetic expressions.

GEOMETRY: Straight lines, parallel lines, Angle properties of triangles, quadrilaterals and polygons.

SUB JUNIOR:

(Kaprekar Contest - VII and VIII Standards)

All the topics for primary and,

ARITHMETIC: Square roots and cube roots, Allegation, average, time and work, time and distance, Races, Games of skill, Travelling around a circle.

MENSURATION: Three Dimensional: cubes, cuboids, sphere, cone, cylinder, pyramids.

ALGEBRA: Algebraic equations of degree 1 and degree 2. Algebraic identities, factorization of algebraic expressions, laws of indices, basics of surds.

GEOMETRY: Triangle inequalities, parallelograms, trapezoids, Pythagoras theorem.

NUMBER THEORY: Prime and composite numbers, divisibility.

[Note: Apart from these, the children shall practice to solve logical questions in basic mathematics].

JUNIOR:

(Bhaskara Contest - IX and X Standards)

All the topics for Primary, Sub junior and the following:

ALGEBRA: Quadratic and Higher degree Algebraic equations, Remainder theorem, Logarithms, Sequences and series, Scales of notations, Mathematical Induction, Basic inequalities.

GEOMETRY: Circle theorems, Chords, Arcs, Angles in segments, Cyclic Quadrilaterals, Tangents, Alternate Segment theorem, intersecting chord theorem, Apollonius theorem, and Stewart's theorem.

NUMBER THEORY: Modular arithmetic, Greatest Integer function, least integer functions.

COMBINATORICS: Fundamental principle of counting, Basics of permutations and combinations, Principle of inclusion and exclusion, Pigeon hole principle.

INTER:

(Ramanujan Contest - XI and XII Standards)

All the topics for Primary, Sub Junior, Junior and the following:

ALGEBRA: Polynomials, inequalities, (C-S inequality), Functional equations.

GEOMETRY: Trigonometric, vector, Coordinate Geometric and Complex number methods may be used.

NUMBER THEORY: Fermat and Wilson theorem, Diophantine equations.

COMBINATORICS: Counting techniques, Recurrence relations.